Digital Audio-Video Signal Processing

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March, 6th 2008

1 Question - 8 pts

Illustrate the Lloyd-Max algorithm for scalar quantization, assuming full knowledge of the source pdf. How does the rate-distortion performance of Lloyd-Max compare with other conventional quantizers (uniform, ECSQ, VQ)?
Explain how the Lloyd-Max quantizer can be designed when the pdf of the source is unknown but a training set of N samples is given. Write the pseudocode of the algorithm in this case.

2 Question - 6 pts

Draw the block diagram of an analysis-by-synthesis speech codec, explaining all functional modules.

3 Question - 6 pts

Draw the block diagram of a motion-compensated video codec, explaining all functional modules. Which are the most relevant coding tools introduced in the new standard H.264/AVC that enable the increase in coding efficiency with respect to previous standards?

4 Question - 12 pts

Consider the `rice` image (`I =imread('rice.png');`).
Implement a baseline version of the JPEG compression algorithm with the following simplifications:

- Luminance component only.
- No DPCM coding of DC coefficient.
- No entropy coding.

The quantization matrix is given by (also available in the attached file `Qjpeg.mat`):

$$\begin{bmatrix} 16 & 11 & 10 & 16 & 24 & 40 & 51 & 61 \\ 12 & 12 & 14 & 19 & 26 & 58 & 60 & 55 \\ 14 & 13 & 16 & 24 & 40 & 57 & 69 & 56 \\ 14 & 17 & 22 & 29 & 51 & 87 & 80 & 62 \\ 18 & 22 & 37 & 56 & 68 & 109 & 103 & 77 \\ 24 & 35 & 55 & 64 & 81 & 104 & 113 & 92 \\ 49 & 64 & 78 & 87 & 103 & 121 & 120 & 101 \\ 72 & 92 & 95 & 98 & 112 & 100 & 103 & 99 \end{bmatrix}$$

Compute the rate-distortion curve for the `rice` image by varying the quantization parameter (plot PSNR vs. rate [bpp]). Since you are not asked to perform entropy coding, you don't know the rate needed for a given quantization parameter. You can estimate the rate using the ρ-domain model:

$$R = \theta(1 - \rho) \tag{1}$$

where ρ is the fraction of *quantized* DCT coefficients equal to zero and θ is an image-dependent parameter (assume $\theta = 5$ in this case).
Now, consider a second image obtained by corrupting the first one with additive white Gaussian noise with standard deviation $\sigma_z = 100$. Compute the rate-distortion curve for this case. Compare the two rate distortion curves and comment the results.